

Gabriel Kassie

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EDUCATION

B.S.E., Computer Engineering

Arizona State University, Tempe, AZ

Barrett, the Honors College

In Progress Coursework: Data Structures and Algorithms, Design and Synthesis of Digital Hardware, Signals and Systems I

Relevant Coursework: Circuits I, Embedded Microprocessor Systems, Object-Oriented Programming and Data Structures

May 2028

3.64 GPA

SKILLS

Hardware and Embedded: KiCAD, OnShape CAD, LTspice, ARM Cortex-M0+ (NXP FRDM-KL46Z), RP2350-CAN

Programming Languages: C, C++, Assembly (MIPS), Python, MATLAB, Java

Web Development and Tools: HTML, CSS, JavaScript, Django, MySQL, Unix/Linux, Git/GitHub, VS Code, Microsoft Office

RESEARCH AND ACADEMIC PROJECTS

Circuits I: Automated Solar Panel Power Meter Lab Project

January 2026 – May 2026

- Produced a portable diagnostic meter prototype for 6V photovoltaics by interfacing a microcontroller with an LCD display to automate real-time voltage, current, and wattage monitoring
- Executed signal conditioning via a non-inverting rail-to-rail Operational Amplifier and a 50Ω shunt resistor to amplify low-magnitude analog signals for high-resolution 10-bit ADC sampling
- Resolved hardware discrepancies between LTspice/TinkerCAD simulations and physical breadboard results by implementing calibration offsets and first-order RC transient analysis to manage discharge timing

Embedded Microprocessor Systems: Autonomous Robot Lab Project

January 2026 – May 2026

- Accomplished 100% mission success and environmental adaptation across multiple surfaces by constructing a bare-metal vehicle platform on the ARM Cortex-M0+ (FRDM-KL46Z) utilizing a dual-motor chassis and a servo
- Navigated real-time trajectories and obstacle avoidance within 2cm path accuracy by programming a PID control system and logic to interface with RGB color, line, and ultrasonic sensors via I2C, ADC, and PWM
- Verified hardware-software reliability and signal integrity by implementing resistor-based voltage scaling for processor protection and utilizing oscilloscopes to troubleshoot sub-millisecond pulse timing

Solar Devils, ASU Solar Car Team: Electrical Integrations Engineer

August 2024 – Present

- Architecting C-based firmware for the RP2350-CAN board to implement safety-critical throttle control logic
- Optimized vehicle-wide electrical systems by designing comprehensive electrical schematics in KiCAD to integrate power distribution and communication protocols
- Enhanced real-time vehicle diagnostics by decoding 700+ CAN Bus data points utilizing Python and Kivy to develop a driver-facing dashboard for critical system diagnostics

Engineering Projects in Community Service, ASU: Prototyping Lead

January 2025 – May 2025

- Developed an Arduino-based system to monitor pH, dissolved oxygen, and ORP levels, applying manual calibration formulas to improve sensor accuracy by 20% for the Vietnam Shrimp Farming project
- Designed and 3D printed a Minecraft Creeper prototype in Onshape CAD, utilizing parametric sketches, extrude operations, and reference planes to meet strict 2.5"x2.5"x6" fabrication constraints

WORK EXPERIENCE

Destination Express LLC, Phoenix, AZ: Software Engineer

December 2024 – January 2025

- Streamlined auto parts delivery logistics for 60+ users by managing 1000+ orders through the deployment of a full-stack web application with Python, Django, and MySQL
- Developed and integrated multi-level dashboards for shippers, drivers, and dispatchers using HTML, CSS, and JavaScript, incorporating secure authentication, real-time order tracking, and management
- Built comprehensive customer and courier features, including new order forms, manifest generation with print functionality, and request handling for quotes, cancellations, and courier applications